

Applications of Laser

- 1. Light Detection and Ranging (LiDAR)**
- 2. Barcode Reader**
- 3. Application in metal work**

Light Detection And Ranging (LIDAR)

LIDAR (Light Detection And Ranging also LADAR) is an optical remote sensing technology that can measure the distance of the target or the other properties of the target by illuminating the target with **light** using pulses from **laser**.

Principle of Working:

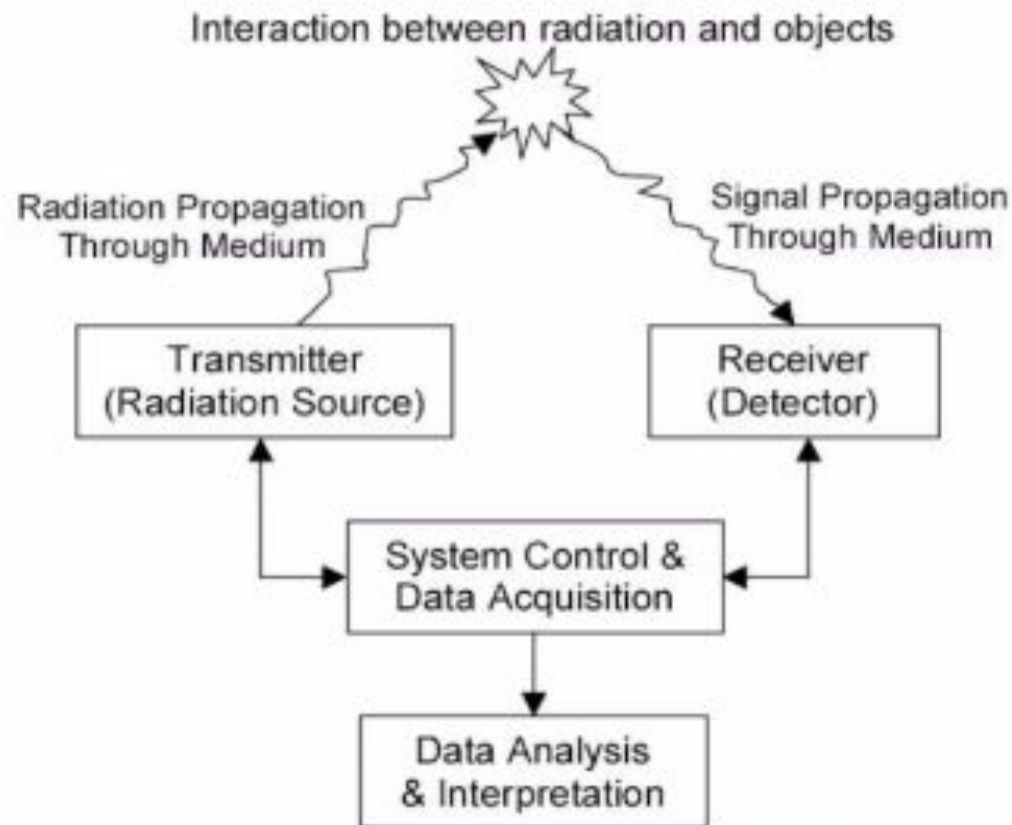
- **The principle is similar as RADAR system used in Navigation Process.**
- **In this technology a Laser source generates the pulses of light which gets reflected from the objects to be scanned.**
- **The reflected laser beam is received by the system receiver which consists of high speed counter circuit to measure the time difference between the incident and received pulses.**
- **Using the measured value of time, the range OR the distance of the target is calculated.**

Working of LIDAR

- 1. Laser Emission:** A LiDAR system emits rapid pulses of laser light towards the target area.
These are in typically In near- infrared region. But different laser system are used as per the applications.
- 2. Reflection and Detection:** When these laser pulses strike an object or surface,
they are reflected back to the LiDAR sensor. The time it takes for the pulses to return is measured.
- 3. Distance Calculated:** By calculating the time it takes for the laser pulses to travel to the object and back,
the system determines the distance using the formula $d = c.t/2$
- 4. Data Collection:** The system collects thousands to million of distances per seconds creating
a dense and detailed 3D map of the surveyed area.
- 5. Data Processing:** The collected data is processed to create detailed 3D representation,
or point clouds which can then be used for various applications.

Working Principle

Physical Picture in LIDAR Remote Sensing



Components used in Lidar

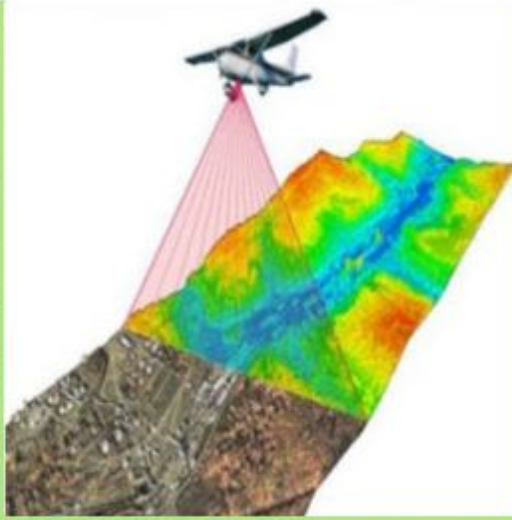
1. Laser
2. Scanner and Optics
3. Photo-detector and receiver electronics
4. Position and navigation systems

LASER

- The Lasers are categorized by their wavelength. Airborne Light Detection and Ranging systems use 1064nm diode-pumped Nd: YAG lasers whereas Bathymetric systems use 532nm double diode-pumped Nd: YAG lasers which penetrate into the water with less attenuation than the airborne system (1064nm). Better resolution can be attained with shorter pulses provided the receiver detector and electronics have sufficient bandwidth to manage the increased data flow.



Ground based LiDAR

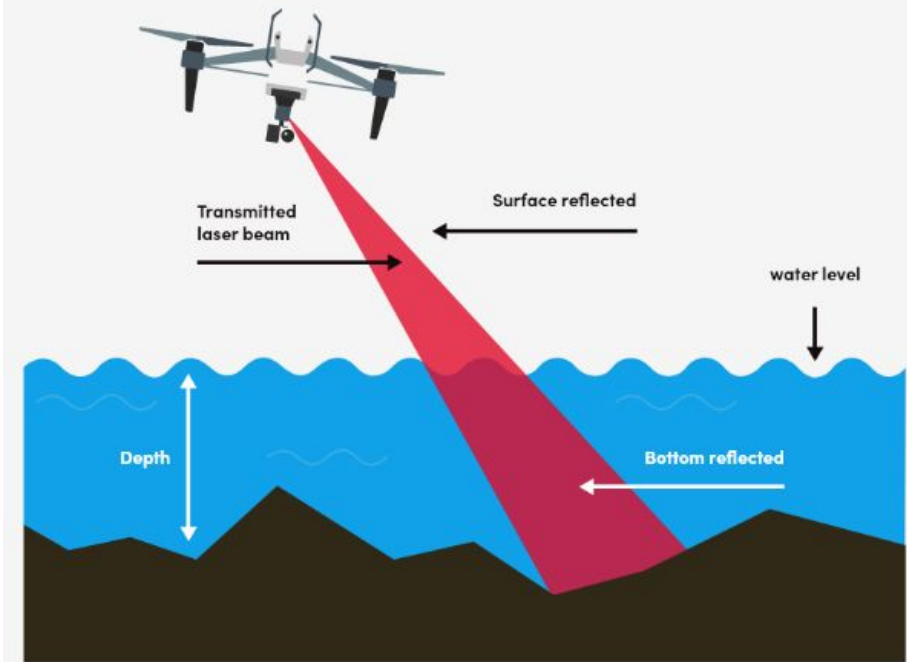


Airborne LiDAR



Spaceborne LiDAR

How Bathymetric LiDAR works?



SCANNER AND OPTICS

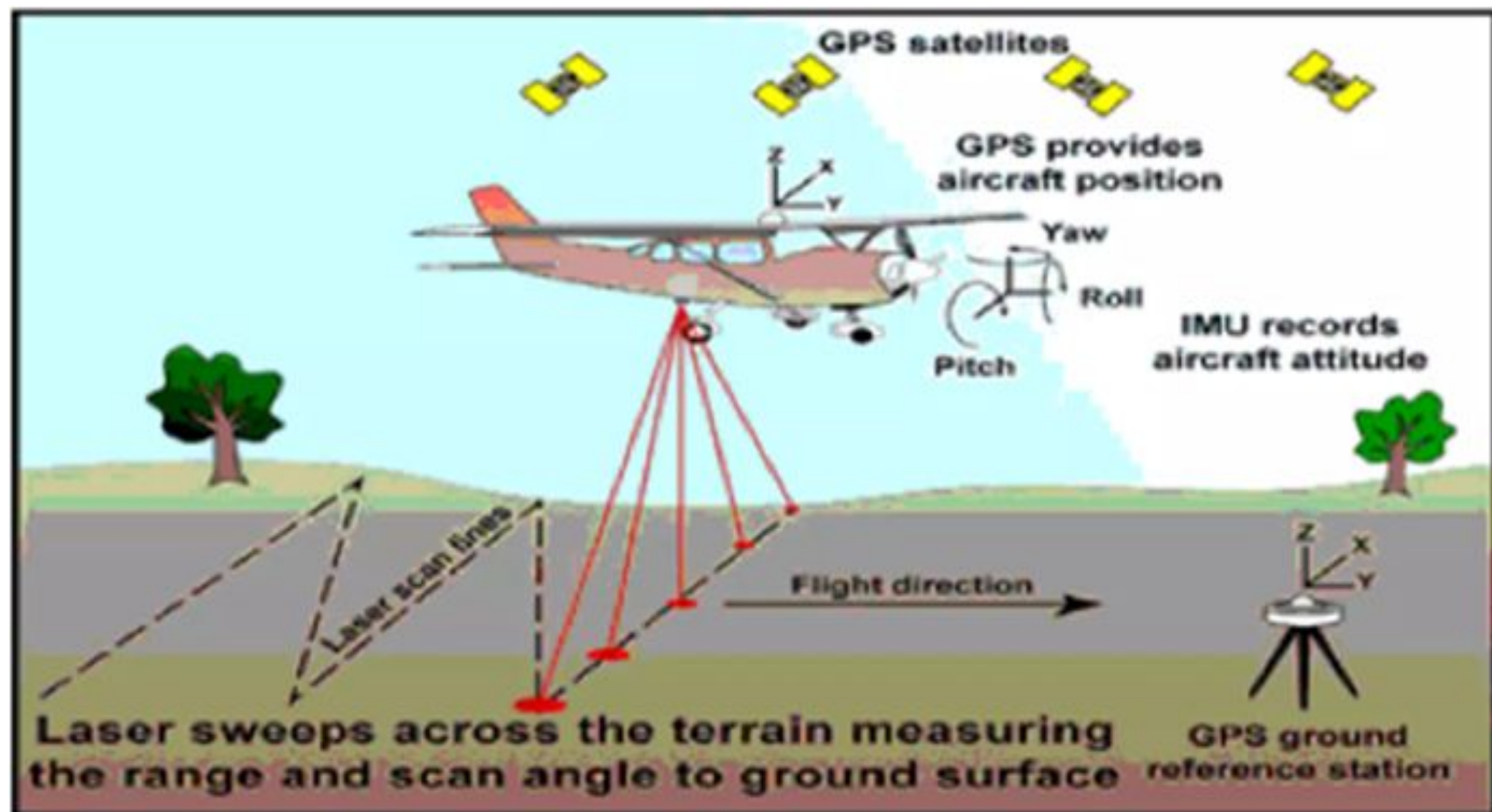
- The speed at which images can be developed is affected by the speed at which it can be scanned into the system. A variety of scanning methods is available for different resolutions such as azimuth and elevation. dual axis scanner, dual oscillating plane mirrors, and polygonal mirrors. The type of optic determines the range and resolution that can be detected by a system.

PHOTO DETECTORS

- The photo detector is a device that reads and records the backscattered signal to the system. There are two main types of photo detector technologies, solid state detectors, such as silicon avalanche photodiodes and photomultipliers

POSITION AND NAVIGATION SYSTEM

- Lidar sensors that are mounted on mobile platforms such as airplanes or satellites require instrumentation to determine the absolute position and orientation of the sensor. Such devices generally include a **Global Positioning System** receiver and an **Inertial Measurement Unit(IMU)**.



Applications of LIDAR

Topographic mapping :

Creating highly accurate topographic maps for various applications, including flood risks, assessment, Forestry, and urban Planning.

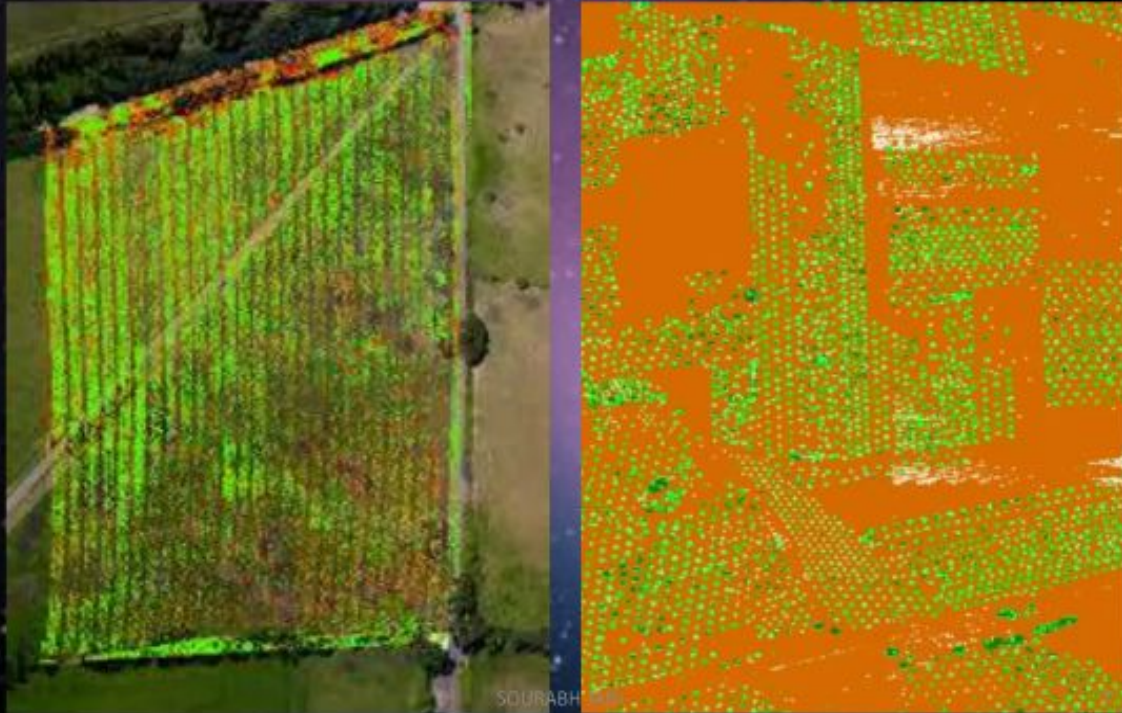
Coastal and Ocean Mapping:

Measuring coastal erosion, mapping underwater topography and assessing changes in coastal environment.

Autonomous Vehicles:

Providing real time 3D maps of the environment for navigation and obstacle avoidance.

Lidar view of Agricultural Land



Aerial Scanning



Terrestrial Scanning



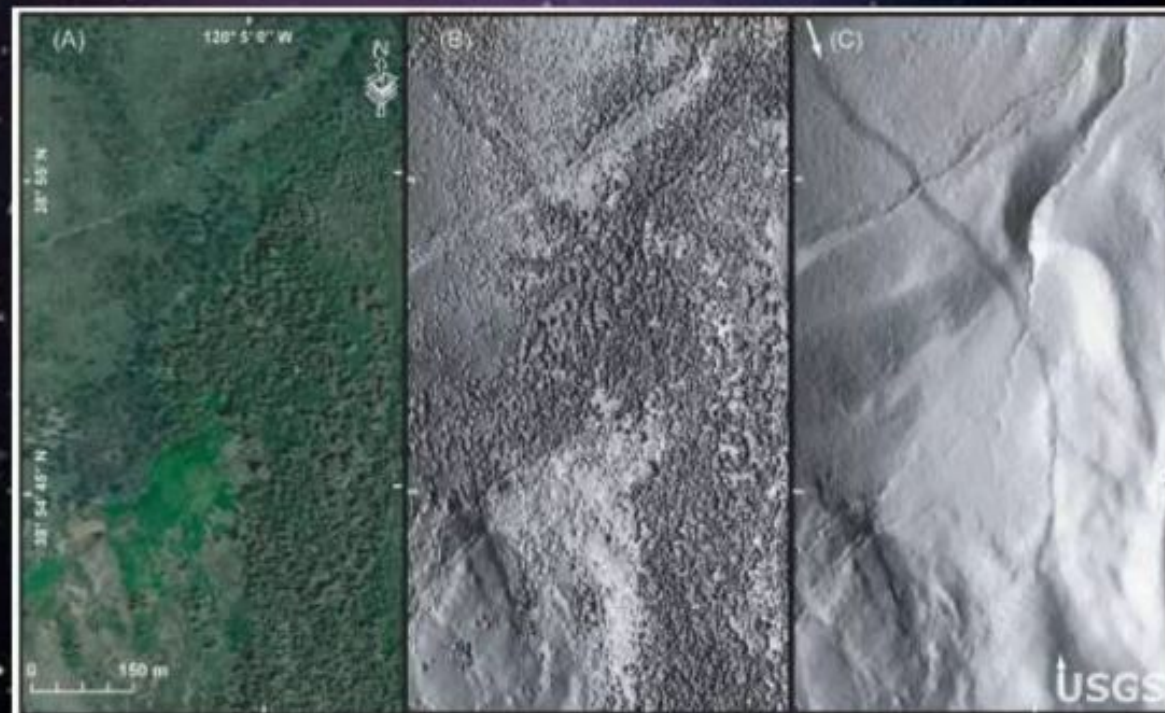
Autonomous Vehicle



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LiDAR Technology Reveals Faults



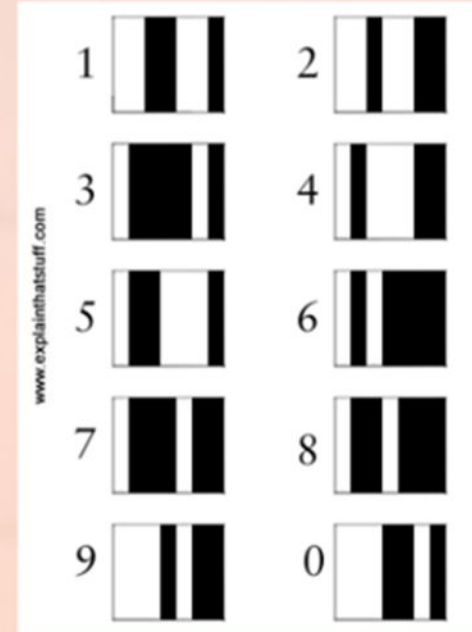
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Barcode Reader

What are Barcodes?

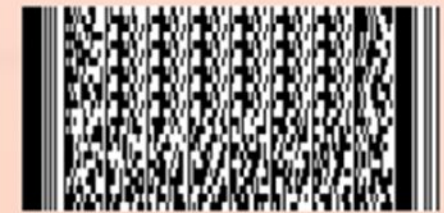
- A barcode is pattern of black and white bars of varying widths used to represent a specific set of numbers that identify products.
- There are two types of barcodes:
 - Linear
 - can hold any type of text (written) information
 - The most common one that we see is called the UPC or the Universal Product Code
 - It is made up of two parts: the barcode and a 12 digit number.
 - 2D
 - type of barcode that is less common and it can store more complex information, such as price, quantity, web address or image



Barcode Patterns



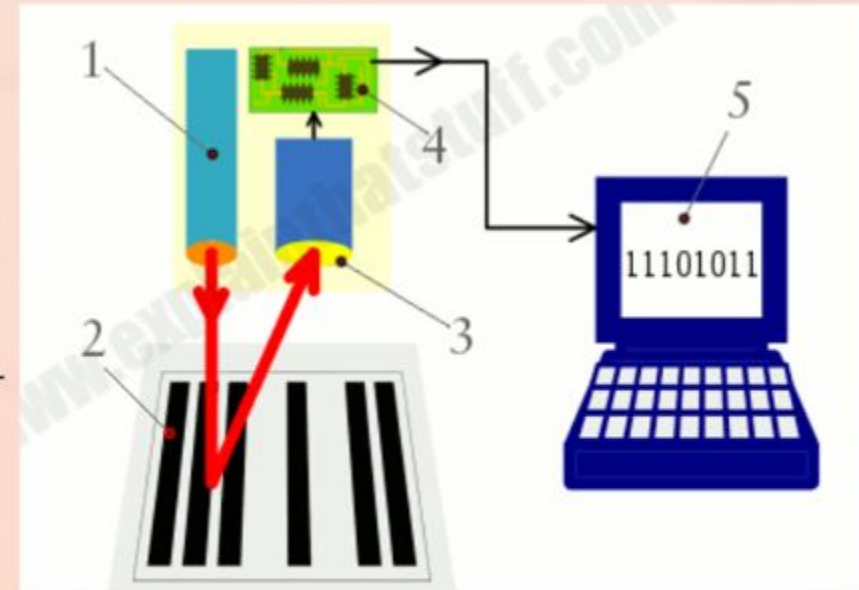
Linear Barcode



2D Barcode

How does a Barcode Scanner Work?

- the scanner emits an LED light or a laser onto the barcode.
- A light is reflected off an optical lens into a light-detecting electronic component called a photoelectric cell.
- The white areas reflect the least light while the black reflect the most.
- The difference in light reflected allows the barcode to identify the unique set of numbers
- As the scanner moves along the barcode, the cell generates a pattern of on-off pulses that correspond to the black and white lines.
- A black line generates an "off" pulse and a white line generates an "on" pulse.
- An electric circuit within the scanner converts these pulses to binary digits (zero or one)
- These digits are sent to the tracking computer which detects those numbers and stores the information contained within the code



What is Barcode Reader??

- A **barcode reader** (or **barcode scanner**) is an electronic device for reading printed barcodes.
- Like a flatbed scanner, it consists of a light source, a lens and a light sensor translating optical impulses into electrical ones.
- Additionally, nearly all barcode readers contain *decoder* circuitry analyzing the barcode's image data provided by the sensor and sending the barcode's content to the scanner's output port.



Types of Barcode Reader

- There are five basic types of barcode readers..
 - pen (wands) type reader
 - slot scanners
 - charge-couple device (CCD) scanners
 - image scanner /camera based reader
 - laser scanners



Pen type readers and Laser Scanners

- Pen type readers consist of a light source and a photo diode that are placed next to each other in the tip of a pen or wand.
- To read a bar code, you drag the tip of the pen across all the bars in a steady even motion.



Slot Scanner

- A slot scanner remains stationary and the item with the bar code on it is pulled by hand through the slot. Slot scanners are typically used to scan barcodes on identification cards.



Charge-Couple Device (CCD) Readers



- A CCD scanner has a better read-range than the pen wand and is often used in retail sales. Typically, a CCD scanner has a 'gun' type interface and has to be held no more than one inch from the bar code. Each time the bar code is scanned, several readings are taken to reduce the possibility of errors.
- A disadvantage of the CCD scanner is that it cannot read a bar code that is wider than its input face.

Camera based Readers Img Scanner



- The newest type of bar code reader currently available are camera-based readers that use a small video camera to capture an image of a bar code. The reader then uses sophisticated digital image processing techniques to decode the bar code.
- It can read a bar code from about 3 to 9 inches away and generally costs less than a laser scanner.

Why use Barcodes??

- When barcodes are used in the business process, procedures are automated to increase productivity and reduce human error.
- Whenever there is a need to accurately identify or track something, bar-coding should be used. For example, in a data entry work environment, workers may be required to enter an enormous amount of data into a customer database system.
- Instead of manually typing a customer identification number into a database, if the information is contained in a barcode, a data entry operator may scan it in. This would increase automation and reduce human error.



Benefits of Barcoding

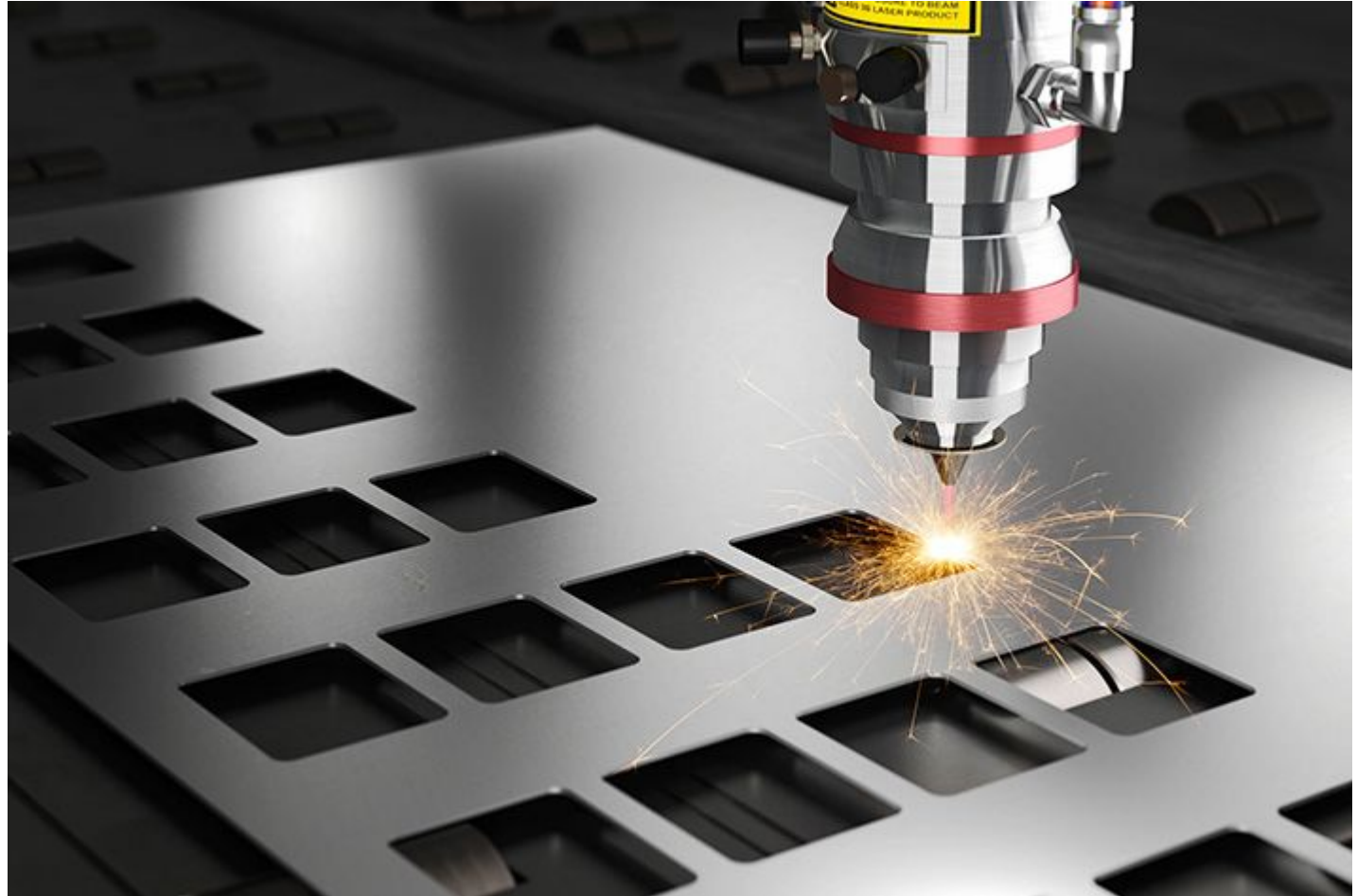
- ACCURACY OF DATA INPUT (ERROR FREE)
- AID EFFECTIVE MANAGEMENT OF RESOURCES AND INVENTORIES
- LABOUR SAVINGS BY AVOIDING MANUAL SYSTEM
- COST EFFICIENT
- REAL TIME DATA COLLECTION
- MEASUREMENT OF WORK IN PROGRESS THROUGHOUT THE FACTORY
- RAPID ACCESS TO TOTAL PRODUCTION COSTS
- MORE ACCURATE DISPATCH

Application in metal work

- 1. Laser Cutting**
- 2. Laser Welding**
- 3. Laser Drilling**
- 4. Laser Marking**
- 5. Laser Surface Treatment**
- 6. Laser Trimming and Ablation**
- 7. Laser Ablation**

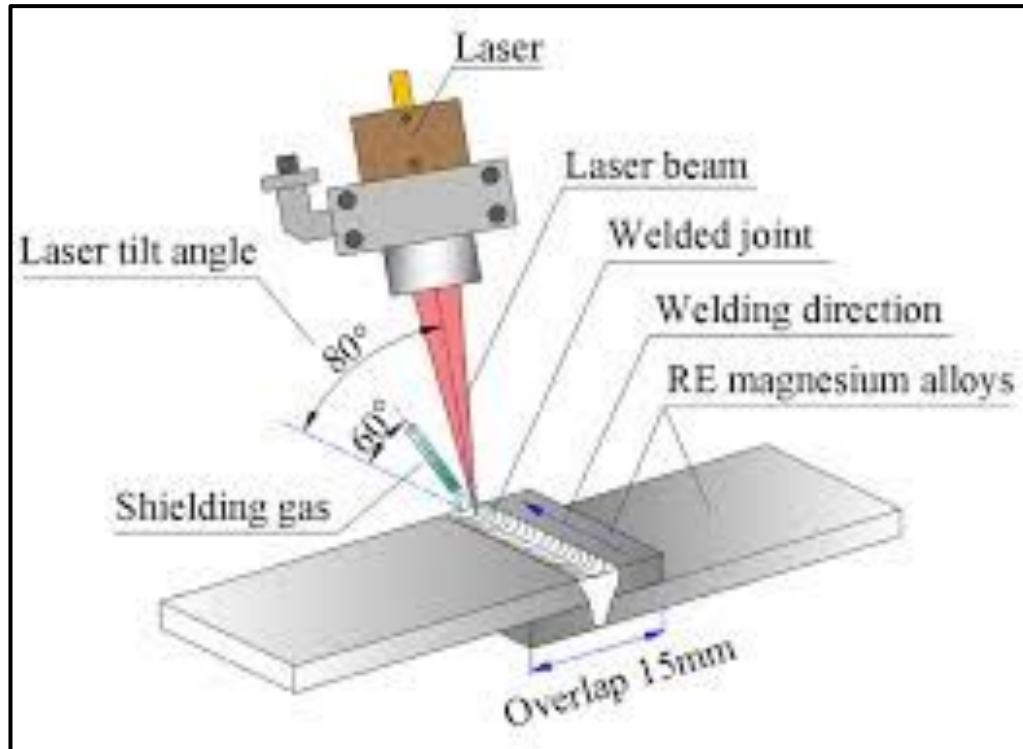
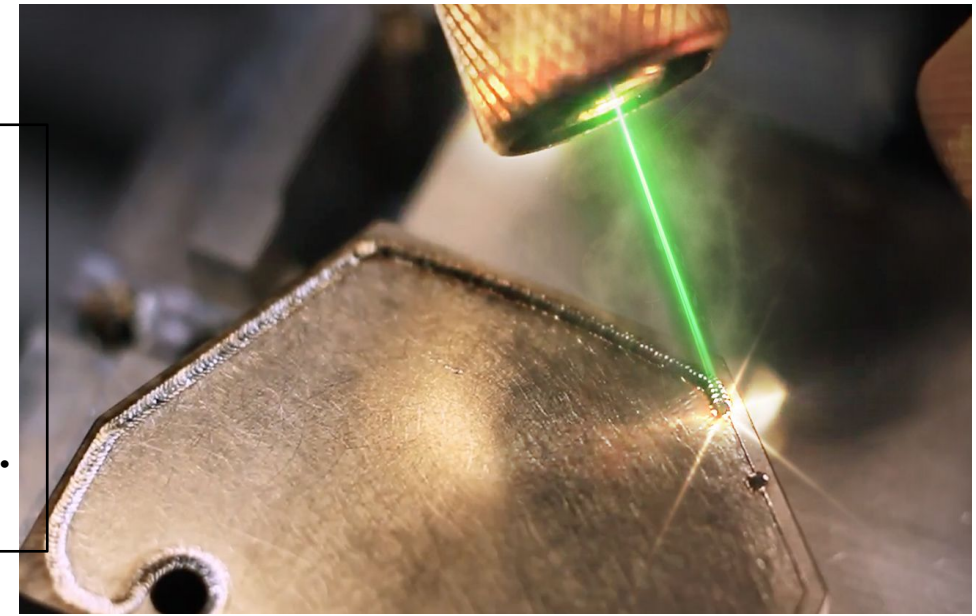
Laser Cutting

- It involves a high-powered laser beam for producing precise cuts.
- Process is controlled by computerized system for accuracy and repeatability.
- Compared to traditional cutting methods laser cutting produces minimal waste.
- It is very much suitable for wide range of metals.



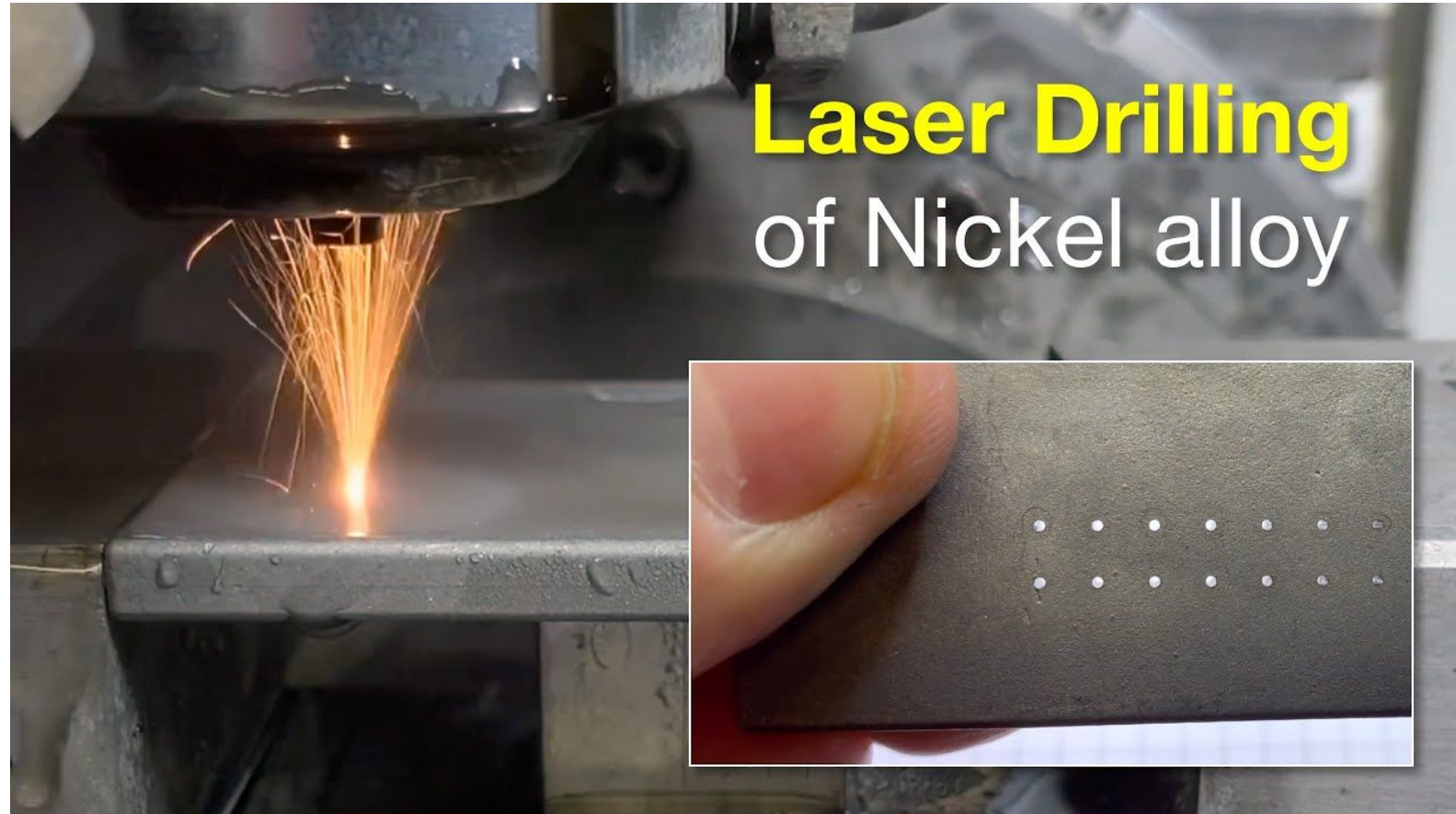
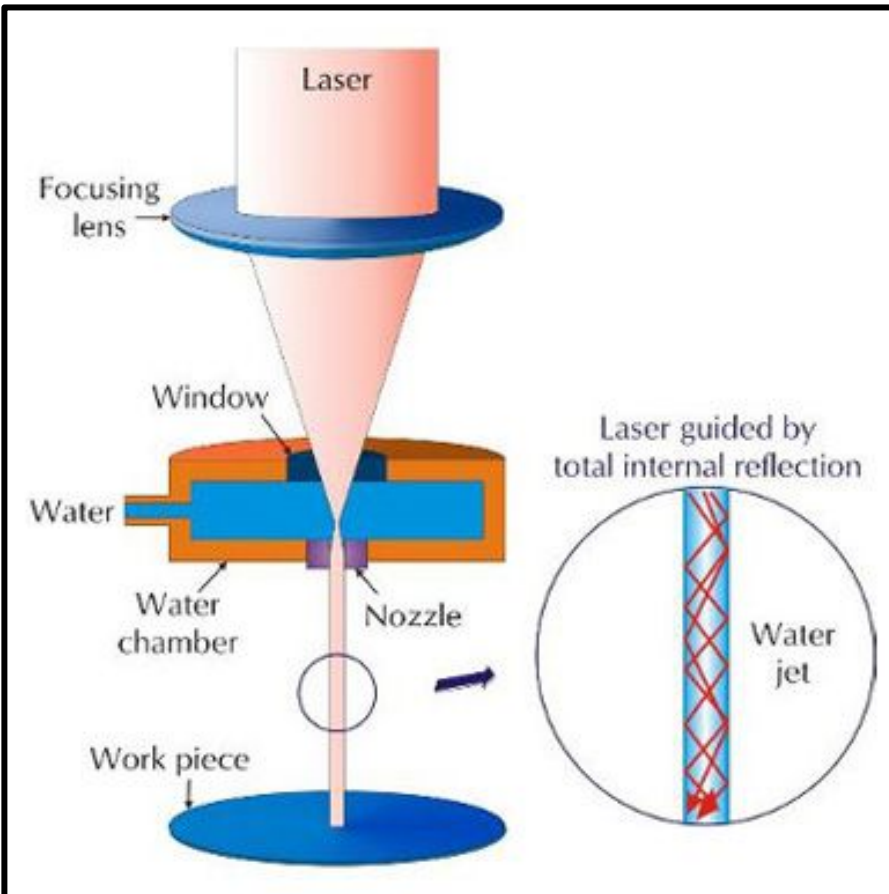
Laser Welding

- ❑ It uses a concentrated laser beam to join metal pieces by melting their edges and joining them together.
- ❑ It can be used for spot welding and continuous welding.
- ❑ Allows for precise control over the heat input, reducing the risk of distortion.
- ❑ It is faster than traditional welding method, especially for thin metals.
- ❑ It is capable of welding hard to reach areas and complex geometries.



Laser Drilling

- A focused laser beam drills precise holes of various sizes and depths in metal sheets.
- It is a precise and efficient process used in metal working to create holes with high depth to diameter ratio, which is difficult using traditional methods.
- The laser does not physically touch the material therefore it has minimal wear and tear and minimum wastage.



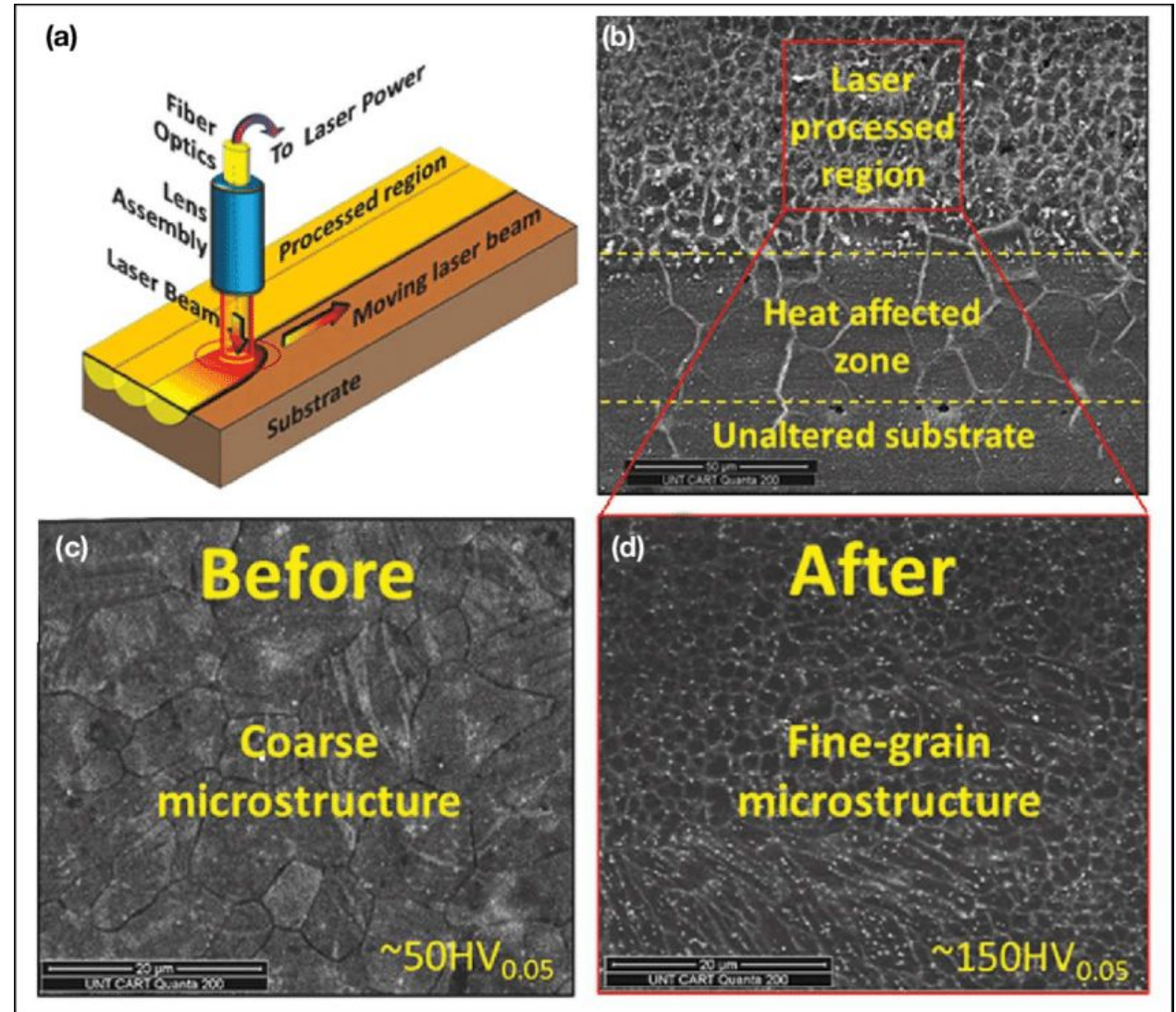
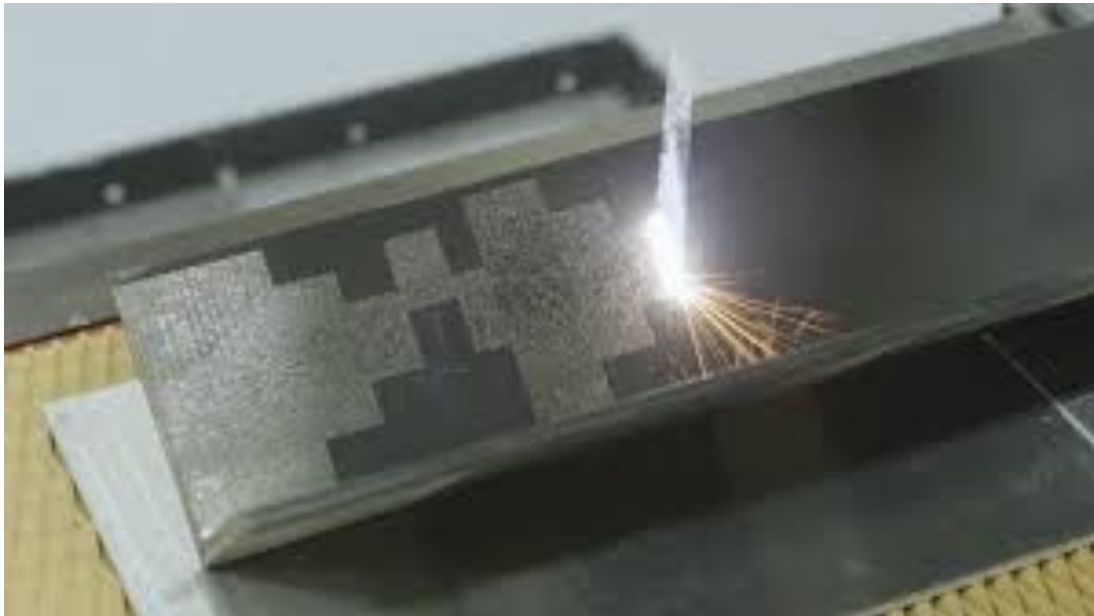
Laser Marking

- ✓ It involves creating permanent marks such as text, logos and serial numbers using laser beam.
- ✓ This is a non-contact clean process suitable for marking various types of metals stainless steel, titanium, and aluminum.
- ✓ Laser produces high resolution marks with intricate details.



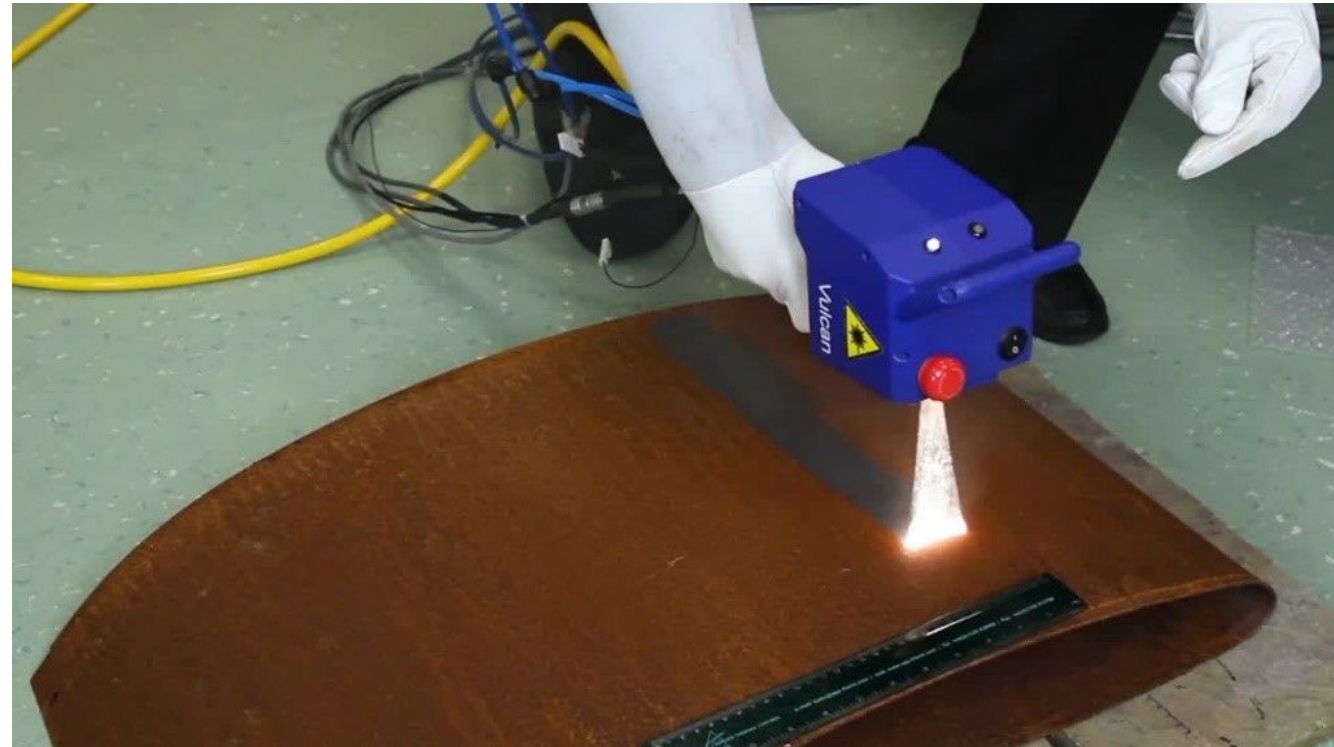
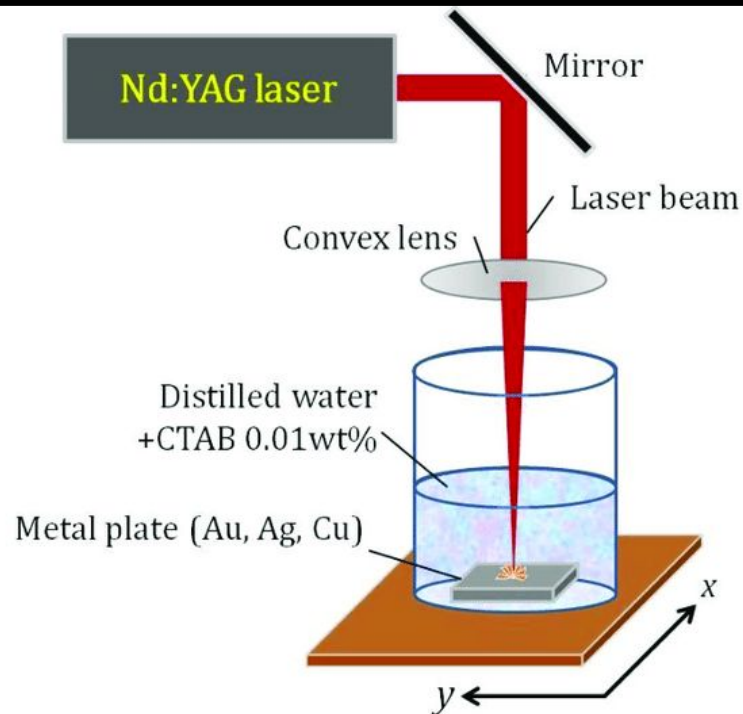
Laser Surface Treatment

- A laser source is used to modify the properties of a metal surface such as hardness, wear resistance, and corrosion resistance.
- This technique includes laser hardening, laser cladding and laser texturing.
- It is a localized treatment without affecting the bulk properties of the material.



Laser Ablation

- ❑ Laser ablation is a material removal process where a laser beam removes material from the surface of a solid by vaporization or sublimation.
- ❑ It is used for a wide range of applications, including cleaning, structuring, and surface modification.
- ❑ It removes the material with minimal thermal impact on the surrounding areas.
- ❑ Applicable to a wide range of materials such as metals, ceramics, polymers, and composites.



Laser Trimming

- Laser beam can be used for precise removal of unwanted material from a metal work piece.
- It is widely used in the electronic industry to fine-tune circuit components for accurate performances.
- It is capable of achieving extremely fine values of components without any mechanical damage.

